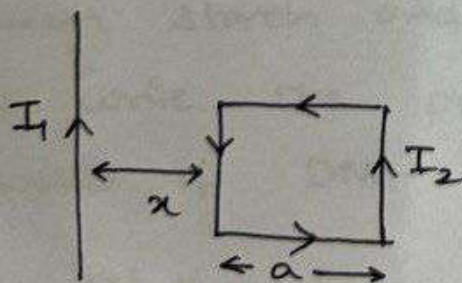


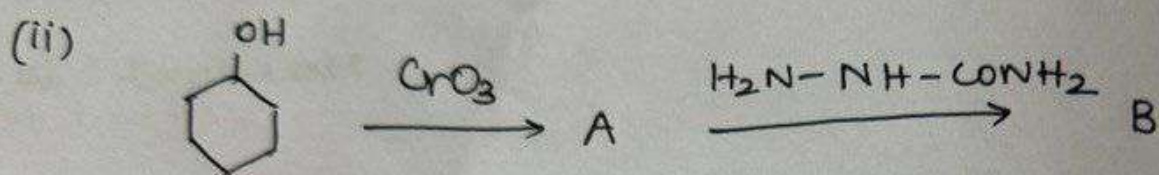
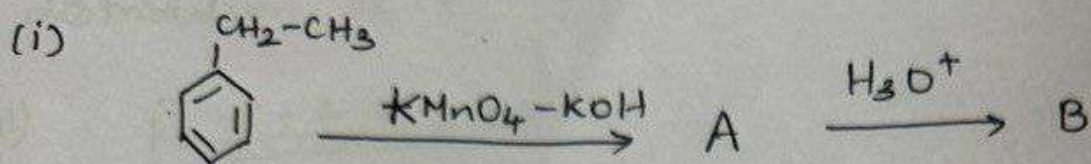
- ① Calculate the de-broglie wavelength associated with an electron revolving in the first excited state of hydrogen atom. The ground state energy of hydrogen atom is -13.6 eV .
- ② Calculate the radius of curvature of an equiconcave lens of refractive index 1.5 , when it is kept in a medium of refractive index 1.4 , to have a power of -50 .
- ③ Why is wave theory of electromagnetic radiation not able to explain photoelectric effect? How does the photon picture resolve this problem?
- ④ A square loop of side ' a ' carrying a current I_2 is kept at a distance x from an infinitely long straight wire carrying a current I_1 , obtain the expression for the resultant force acting on the loop.



- ⑤ An aeroplane is flying horizontally from west to east with velocity 900 km/h . Calculate the potential difference developed between the ends of its wings having a span of 20 m . The horizontal

component of earth's magnetic field is $5 \times 10^{-4} \text{ T}$ and the angle of dip is 30° .

⑥ write structures of the compound A and B in the following :-

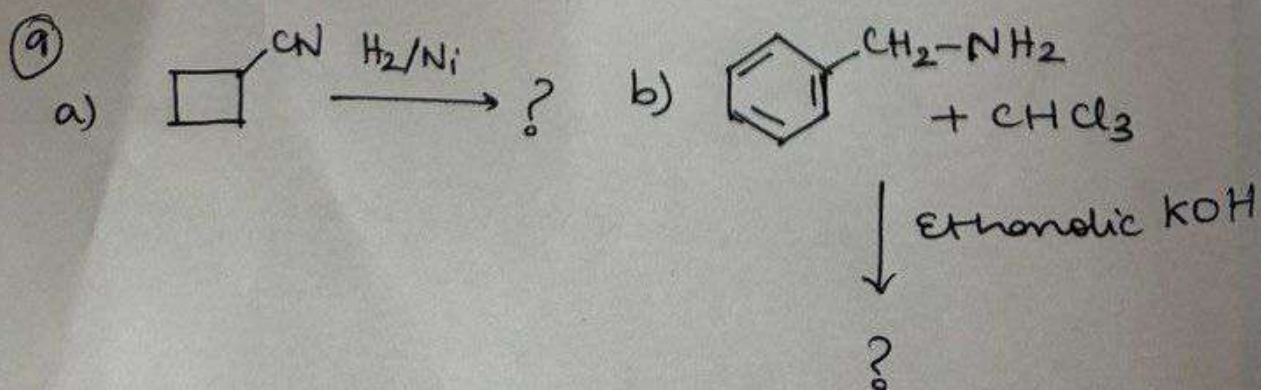


⑦ Out of $[\text{CoF}_6]^{3-}$ and $[\text{Co}(\text{en})_3]^{+3}$, which one of the complex is?

- (i) More stable
- (ii) Paramagnetic
- (iii) Inner orbital complex.
- (iv) High spin complex.

⑧ What is the structural basic difference between starch and ~~cellulose~~ Cellulose?

Also, write the products obtained after hydrolysis of DNA.



10) Two numbers are selected from first six even natural numbers at random without replacement. If x denotes the greater of the two numbers selected. find the probability distribution of x .

11) $\int \frac{x+2}{\sqrt{x^2-4x-5}} dx = ?$

12) Evaluate $\int_{-a}^a f(x) dx$, where $f(x) = \frac{9^x}{1+9^x}$

13) Solve the differential Equation:

$$(y - \sin^2 x) dx + \tan x dy = 0$$

14) The function $f(x) = 2x^3 - 15x^2 + 36x + 6$ is increasing in the interval.

- (A) $(-\infty, 2] \cup (3, \infty)$ (B) $(-\infty, 2)$
 (C) $(-\infty, 2] \cup [3, \infty)$ (D) $[3, \infty)$

15) find the domain of $\cos^{-1}(2x-3)$?

16) Evaluate $\int_1^2 \left[\frac{1}{x} - \frac{1}{2x^2} \right] e^{2x} dx$

17) solve for x :

$$\sin^{-1}(1-x) - 2\sin^{-1}(x) = \pi/2$$